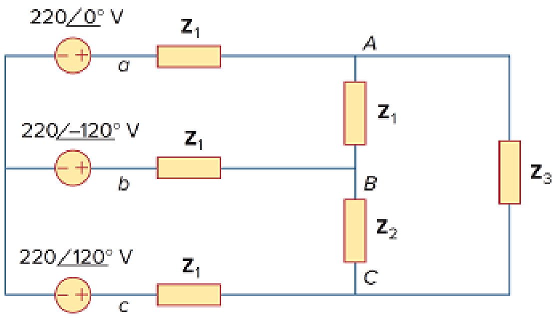
Problem

Use *PSpice or MultiSim* to find currents **I***aA* and **I***AC* in the unbalanced three-phase system shown in Fig. 12.69. Let

$$\mathbf{Z}_l = 2 + j, \qquad \mathbf{Z}_1 = 40 + j20\ \Omega,$$

$$\mathbf{Z}_2 = 50 - j30\ \Omega, \qquad \mathbf{Z}_3 = 25\ \Omega$$

Figure 12.69



Step-by-step solution

Step 1 of 4

Let $\omega = 1$,

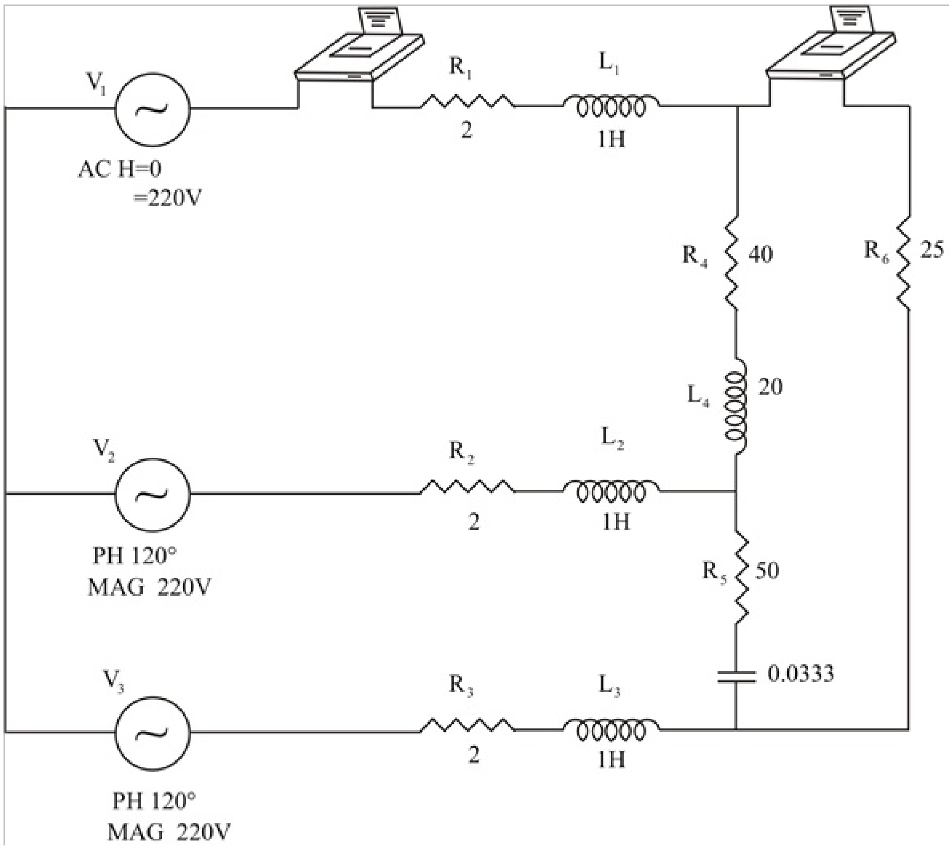
So that,

$$L = \frac{X}{\omega} = 204$$

And,

$$C = \frac{1}{\omega X} = 0.0333\,F$$

Step 2 of 4



Step 3 of 4

Frequency	$1M(V\_PRINT1)$	$IP(V\_PRINT1)$
$1.592E-01$	$1.867E+01$	$1.589E+02$
Frequency	$1M(V\_PRINT2)$	$IP(V\_PRINT2)$
$1.592E-01$	$1.238E+01$	$1.441E+02$

Step 4 of 4

From the output file, the required currents are,

$$I_{aA} = 18.60\angle 158.9^\circ\ A$$

$$I_{AC} = 12.38\angle 144.1^\circ\ A\ .$$